Edouard Barrier

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Academic Employment

Department of Earth and Space Sciences, University of Washington Seattle, WA, USA
08/2026 - Present

Advisors: Profs. David Catling and Joshua Krissansen-Totton

Working on:

- Coupled atmosphere-interior planet evolution
- Redox Biosignatures

Education

Institute of Astronomy, University of Cambridge; Darwin College Cambridge, UK
PhD, Astronomy 10/2022 - 06/2026

Advisor: Prof. Nikku Madhusudhan

Thesis Title: 'Climate and Atmospheric Dynamics of temperate Sub-Neptunes'

- A new generalised GCM for temperate sub-Neptune exoplanets, building on the ExoCAM framework.
- A new convection parametrisation applicable to diverse modes of convection in the sub-Neptune regime.
- First GCM study of Hycean worlds: analysis of atmospheric dynamics, first global GCM study of moist convective inhibition, updated haze-albedo constraints on K2-18 b allow for it to be Hycean.
- Sea ice as an environment for prebiotic chemistry and abiogenesis on sub-Neptunes, investigation of sea ice formation and meteoritic impactor fragments as mechanism for concentrating feedstock molecules.
- Habitable zones more broadly, self-consistent hazes, mini-Neptune GCM intercomparison with the CAMEMBERT collaboration.

Gonville & Caius College, University of Cambridge Cambridge, UK
MSc, Astrophysics 10/2021 - 06/2022

Grade: 1st Class, and 1st place in cohort
Masters' Thesis with Prof. Cathie Clarke, 'Spectro-astrometric signatures of disk winds'
Courses in Exoplanets, Astrophysical fluid dynamics, Disks, Stars, Geophysics

BA (Hons), Astrophysics 10/2018 - 06/2021

Grade: 1st Class, and 1st place in cohort
Courses in Astrophysics, Physics, Earth Science, Computer Science, Maths

Publications

- **Barrier**, Madhusudhan, and Rigby (2026), 'Origin of Life locations for Hycean and Ocean planets', *MNRAS*
- **Barrier** and Madhusudhan (2025), 'General Circulation Models of Hycean Worlds', *MNRAS*
- **Barrier** and Madhusudhan (2025), 'A new convection scheme for GCMs of temperate sub-Neptunes', *MNRAS*
- Madhusudhan, Moses, Rigby, and **Barrier** (2023), 'Chemical Conditions on Hycean Worlds', *Faraday Discussions*

Talks and Posters

Seminar, UW Astrobiology — Seattle, WA, USA
Climates of Temperate Sub-Neptunes — 10/2026

Seminar, Bowie Seminar Series — Virtual
Climates of Temperate Sub-Neptunes — 10/2026

Talk, Exoplanets VI — Porto, Portugal
The 3D Runaway Greenhouse Effect on Temperate Sub-Neptunes — 07/2026

Poster, AbSciCon — Madison, WI, USA
The Runaway Greenhouse Effect on Temperate Sub-Neptunes — 05/2026

Talk, LCLU Coffee Talk Series — Cambridge, UK
Sea Ice as an Origin Of Life Location for Ocean Worlds — 01/2026

Poster, Buffet-5 Meeting — Bordeaux, France
Parametrising Convection in sub-Neptune Atmospheres — 09/2025

Seminar, Wisconsin Centre for Origins Research — UW-Madison, Madison, WI, USA
GCM Simulations of Temperate Sub-Neptunes — 07/2025

Talk, ExoClimes VII — Montreal, Canada
Runaway Greenhouse Thresholds on K2-18 b — 07/2025

Talk, UKExom 2025 — Leeds, UK
Atmospheric dynamics and climates of temperate Sub-Neptunes — 04/2025

Talk, Leverhulme Centre for Life in the Universe Science Day — Cambridge, UK
Atmospheric dynamics and climates of temperate Sub-Neptunes — 03/2025

Poster, Origins Federation Conference III — Cambridge, UK
A new convection scheme for GCMs of temperate sub-Neptunes — 09/2024

Talk, Leverhulme Centre for Life in the Universe Science Day — Cambridge, UK
A new deep convection model for exoplanet atmospheres — 03/2024

Seminar, Institute of Astronomy — Cambridge, UK
A new deep convection model for exoplanet atmospheres — 03/2024

Awards and funding

- 'Institute of Astronomy Prize' for best performance in Part III Astrophysics, 2022
- 'Duncan Bruce Memorial Prize for Physics' for best Part III result in physical Natural Sciences at Caius, 2022
- College Scholarships for performances in examinations, 2021 and 2022
- Travel funding for Breakthrough Discuss conference (£1200 from Breakthrough initiatives), 2024
- Travel funding for Origins Federation conference III (£1400 from Northeastern University), 2023
- UKRI Studentship for study towards a PhD in Cambridge, 2022-2026

Academic Teaching

- Supervisor of 3rd year undergraduate 'Structure and Evolution of Stars' course. Small group teaching (2-4 people) and marking. (40 hours)
- Teaching Assistant, Masters' Course on Exoplanetary Atmospheres and Interiors. Ran office hours, usually to 5-10 students, and marked work including examination scripts. (60 hours)

Academic Service and Outreach

- Member of the Postgraduate student forum from 2022, key link between students and wider faculty
- Star-gazing lead + general helper at Institute of Astronomy open evenings, attended by an average of 150 people.
- Member of the International Women's Day Committee 2025, helped with organisation of the various events, talks, schools visiting.
- Designed and helped run a Venus-based activity for children at the IOA Open day 2023, attended by 1000+ people.

Selected Work Experience and Volunteering

Corrie Energy London, UK
Head Software and Model Developer 07-10/2021 and 07-10/2022

- Developed Python ray-tracing algorithm to model effectiveness of patented dual-axis solar tracking technology, both for standalone solar panels and arrays. Containerised with Docker, production runs on AWS.
- Guided development of trackers and optimised tracking algorithm and field spacings. This increased power generation by an extra 5 %, allowing the technology to be cost-effective.

Gonville and Caius May Ball Committee 2022 Cambridge, UK
Treasurer 08/2021-08/2022

- Controlled £210,000 budget for 1400 person event.
- Managed ticketing team of 5 and increased sales by 24%.

Cambridge University French Society Cambridge, UK
Co-President 06/2021-06/2022

- Organised events attended by over 800 people.
- Instituted membership system which tripled revenue.

Darwin College Students Association Cambridge, UK
Events Officer 11/2022-11/2024
Organised BBQs, Music nights, sports showings, board games nights

GoToCo Zhaoqing, Guangdong, China
Teacher 07/2019
Taught English and sports to 60 9-11 year old students.

Little House of Science London, UK
Teacher 07/2018-04/2019
Taught at science-themed holiday camps for children 4-12, held at Imperial College London

Skills

Languages	Native French and English, Basic Chinese and Italian
Programming Languages	Python, Fortran, C++, Java, Bash, LaTeX
Software	NumPy, SciPy, Pandas, Git, Copilot, NetCDF
Technical	High-Performance Computing, Data Analysis & Visualisation, AI-assisted software development